

QUANTUM COSMOLOGY, EUCLIDEAN WORMHOLES AND LIOUVILLE THEORY

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Based on [2602.23432] with P. Betzios, O. Papadoulaki, I. Gialamas
and on works to appear with P. Betzios, O. Papadoulaki, I. Tsiaras



I. Wineglass cosmology and wormholes

- No-boundary instantons
- Wineglass Euclidean preparation
- Phase diagram of saddles
- Cosmological correlators

II. Boundary Liouville theory

- Spacelike vs timelike regimes
- Boundary bootstrap data
- Moore–Seiberg consistency
- Preliminary timelike results

Part I – Wineglass cosmology and wormholes

NO-BOUNDARY INSTANTONS AND THE WAVEFUNCTION OF THE UNIVERSE

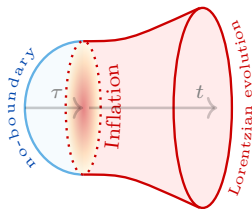
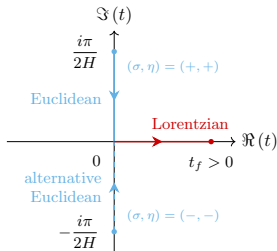
MINISUPERSPACE SADDLE POINT PROBLEM

$$\mathcal{S}_E = -\frac{1}{2} \int d^4x \sqrt{g_E} \left(\frac{\mathcal{R}}{2\kappa} + \Lambda \right) + \mathcal{S}_{\text{GHY}}, \quad \kappa = 8\pi G_N.$$

$$ds^2 = d\tau^2 + a^2(\tau) d\Omega_3^2 \Rightarrow a(\tau) = \frac{\cos(H\tau)}{H}, \quad \Lambda = 3H^2.$$

Four "complex" regular saddles with $a(\tau_i) = 0$, $a(\tau_f) = b$

$$i\mathcal{S}_L^{(\sigma,\eta)} = \sigma \frac{4\pi^2}{\kappa H^2} \left[1 - \eta i((bH)^2 - 1)^{3/2} \right], \quad \sigma, \eta = \pm 1.$$



HARTLE-HAWKING vs VILENKIN PROPOSAL IN SINGLE-FIELD SLOW-ROLL INFLATION

SINGLE-FIELD MINISUPERSPACE SLOW-ROLL

Scalar field minimally coupled to gravity with slow-roll potential $V(\phi) \sim \Lambda = 3H^2$.

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Hartle-Hawking

- 2 instantons
- $\Psi_{\text{HH}}(a, \tilde{\phi})$

$$\propto \exp \left[+ \frac{4\pi^2}{\kappa \tilde{V}} \right] \cos \left[\frac{4\pi^2}{\kappa \tilde{V}} ((a\tilde{V})^2 - 1)^{3/2} - \frac{\pi}{4} \right].$$

- + sign in the exponent $\Rightarrow$ **enhanced probability for small $\tilde{V} \Rightarrow$ small number of e-folds.**
- (But good behavior under perturbations $t = -i\tau$)

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Vilenkin (Tunneling)

- 1 "outgoing" instanton
- $\Psi_{\text{T}}(a, \tilde{\phi})$

$$\propto \exp \left[- \frac{4\pi^2}{\kappa \tilde{V}} \left(1 + i \left((a\tilde{V})^2 - 1 \right)^{3/2} \right) \right] .$$

- **-** sign in the exponent $\Rightarrow$ **enhanced probability for large $\tilde{V} \Rightarrow$ large number of e-folds.**
- (But bad behavior under perturbations $t = +i\tau$)

Goal: keep the good perturbative behavior while avoiding the unwanted weighting.

BASIC MECHANISM

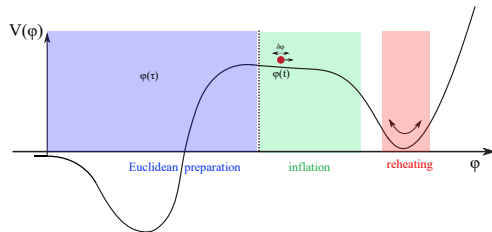
HH gives $\Psi \sim e^{-\mathcal{S}_E}$. Can a negative-potential Euclidean region reverse the problematic weighting?

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BASIC MECHANISM

HH gives $\Psi \sim e^{-\mathcal{S}_E}$. Can a negative-potential Euclidean region reverse the problematic weighting?

- Euclidean solutions are asymptotically EAdS.
- Solutions of interest need to be $\mathbb{Z}_2$ symmetric and present a local maximum $\Rightarrow$ Wineglass Euclidean wormholes.
- Spherical wormholes require negative Euclidean energy density.



CONCRETE REALIZATION IN EINSTEIN+SCALAR+ $U(1)^3$ THEORY

$$\mathcal{S}_E = \int d^4x \sqrt{g_E} \left(-\frac{\mathcal{R}}{2\kappa} + \frac{1}{2}(\partial_\mu \phi)^2 + V(\phi) + \frac{1}{4}F_{\mu\nu}^I F^{\mu\nu,I} \right) + \mathcal{S}_{\text{GHY}} .$$
$$ds^2 = d\tau^2 + a^2(\tau) d\Omega_3^2, \quad \phi = \phi(\tau), \quad A^{(I)}(\tau) = \omega^{(I)} H(\tau) .$$

$$\tilde{\phi}'' + 3 \frac{a' \tilde{\phi}'}{a} - \frac{d\tilde{V}(\tilde{\phi})}{d\tilde{\phi}} = 0 ,$$
$$\frac{a'^2}{a^2} - \frac{1}{a^2} + \left(\tilde{V}(\tilde{\phi}) - \frac{\tilde{\phi}'^2}{2} \right) - \frac{\tilde{\rho}_{\text{rad}}^E}{a^4} = 0 ,$$
$$\tilde{\rho}_{\text{rad}}^E = 2\kappa a^4 \left[\left(\frac{H'}{a} \right)^2 - \frac{4H^2}{a^4} \right] < 0 .$$

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- **Background subtraction** was employed.
- **IR condition:** solutions are even. Then $\Rightarrow Z(H_{\text{UV}}, \phi^{(-)} = 0)$.
- Analyse both Neumann and Dirichlet BCs for the scalar and the gauge field.

A ZOOLOGY OF EUCLIDEAN SOLUTIONS

In practice, for Wineglass running solutions, postulate a scale factor and find the associated potential

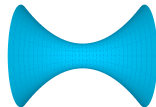
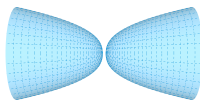
$$a_{WG}^2(\tau) = \begin{cases} \frac{1}{4H_I^2 \cosh(\bar{\tau}^I)} \left(\sqrt{A_I} \cosh(\bar{\tau}^I) - \frac{1}{\sqrt{A_I}} \right)^2, & \bar{\tau}^I = 2H_I(\tau - \tau_{\min}) < 0. \quad (\text{I}) \\ \frac{1}{4H_{II}^2} \left(\cos(\bar{\tau}^{II}) - 16\tilde{\rho}_{\text{rad}}^E H_{II}^2 - 1 \right), & 2H_{II}\tau_{\min} < \bar{\tau}^{II} = 2H_{II}\tau < 0. \quad (\text{II}) \end{cases}$$

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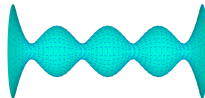
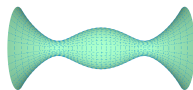
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**Disconnected,
non-running and $F = \star F$**



**Non-running wormhole
and $\tilde{\rho}_{\text{rad}}^E < 0$**

**Running Wineglass $\phi(\tau)$
and $\tilde{\rho}_{\text{rad}}^E < 0$**



Oscillatory Wineglass

RESULTS: DIAGRAM OF DOMINANCE

- Solutions are compared with the same sources at the EAdS boundary.
- Scalar field characterised by $\Delta_+ = 2$, $\Delta_- = 1$. Scalar field source set to zero to compare with non-running solutions.
- **Energy density is an IR parameter!** $\tilde{\rho}_{\text{rad}}^E = -8\kappa H_0^2$.

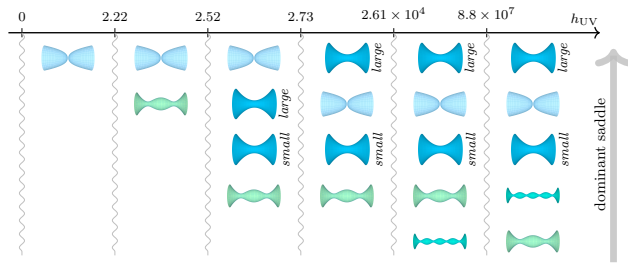


Figure: Diagram of dominance with Neumann boundary conditions for the scalar field and Dirichlet boundary conditions for the gauge field.

COSMOLOGICAL CORRELATORS: MASSLESS SCALAR PROBE IN A FIXED BACKGROUND

Massless scalar probe on top of the background (QFT in curved spacetime)

$$S_\Phi = -\frac{1}{2} \int d^4x \sqrt{-g} \partial_\mu \Phi \partial^\mu \Phi, \quad \Phi = \sum Q_{n\ell m} \Phi_{n\ell m}.$$
$$0 = \ddot{\Phi}_{n\ell m}(t) + 3\frac{\dot{a}}{a} \dot{\Phi}_{n\ell m}(t) + \frac{n^2 - 1}{a^2} \Phi_{n\ell m}(t).$$

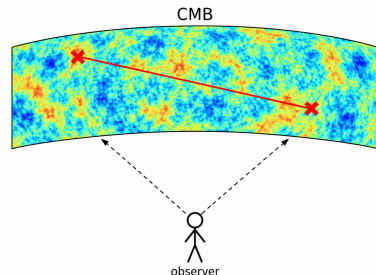
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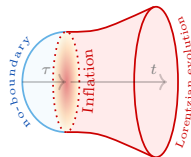
- **Bayesian** question.
- For a **free theory**, $\langle |\Phi_{n\ell m}|^2 \rangle$ is obtained from the classical EOM, and the boundary conditions give the vacuum prescription. We impose *Euclidean normalizability* at the EAdS boundary.



RESULTS: NO-BOUNDARY VS WINEGLASS

No-boundary proposal = pure de Sitter. Exact results are known for the *Euclidean vacuum*.

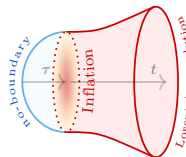
$$\langle |\Phi_{n\ell m}|^2 \rangle = \mathcal{P}_{N.B.}(n; t = \infty) = \frac{H^2}{2n(n^2 - 1)}.$$



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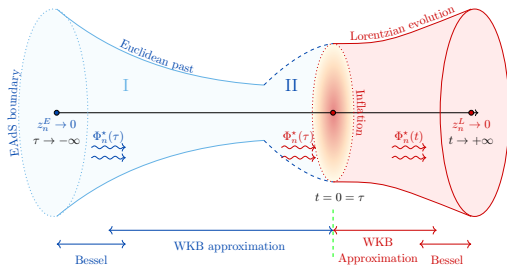
$$\langle |\Phi_{n\ell m}|^2 \rangle = \mathcal{P}_{N.B.}(n; t = \infty) = \frac{H^2}{2n(n^2 - 1)}.$$



- For the *Wineglass Euclidean vacuum*, a WKB matching procedure at large n gives

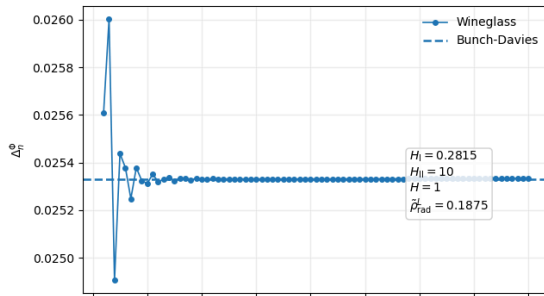
$$\mathcal{P}_{WG}(n; t = \infty) \approx \mathcal{P}_{N.B.}(n; t = \infty).$$

- Also, $\mathcal{P}_{WG}(1; t = \infty) < \infty$: boundary conditions break the shift symmetry.



MAIN CONCLUSION

Wineglass corrections are confined to low harmonics; the spectrum rapidly approaches the Euclidean vacuum result at large n .



$$\text{Dimensionless power spectrum: } \Delta_n = \frac{n(n^2 - 1)}{2\pi^2} \mathcal{P}(n; t = \infty).$$

Part II — Boundary Liouville theory

A complementary exactly solvable setting for quantum gravity

THE ACTION

$$S_L = \frac{1}{4\pi} \int \sqrt{\hat{g}} d^2x \left(\hat{g}^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + Q \hat{R} \phi + 4\pi\mu e^{2b\phi} \right) \quad \Rightarrow_{\text{genus } 0} S_L = \frac{1}{4\pi} \int dz d\bar{z} (\partial_z \phi \partial_{\bar{z}} \phi + 4\pi\mu e^{2b\phi})$$
$$c = 1 + 6Q^2 = 1 + 6(b + b^{-1})^2.$$

$c \geq 25, b \in (0, 1] \Rightarrow$ "Spacelike" regime. $c \leq 1, \hat{b} = -ib \in (0, 1] \Rightarrow$ "Timelike" regime.

BULK LIOUVILLE THEORY

THE ACTION

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AXIOMATIC BOOTSTRAP

Spectrum $\mathcal{H}$ is a rep of the Virasoro algebra closed under OPE. Commutativity of fields (in Euclidean). In Liouville field theory,

- Conformal dimensions $\Delta = \frac{c-1}{24} - P^2 \Rightarrow \mathcal{H} = \int_{i\mathbb{R}+(\epsilon)}^\oplus dP \mathcal{V}_P \otimes \bar{\mathcal{V}}_P \Rightarrow$ **Continuous**.
- Spacelike Liouville is **unitary** but has no **vacuum** (and no degenerate rep in the spectrum)!
- Timelike Liouville is **non-unitary** but has a **vacuum** (and a degenerate rep in the spectrum)!

BOUNDARY LIOUVILLE BOOTSTRAP

ADD A BOUNDARY

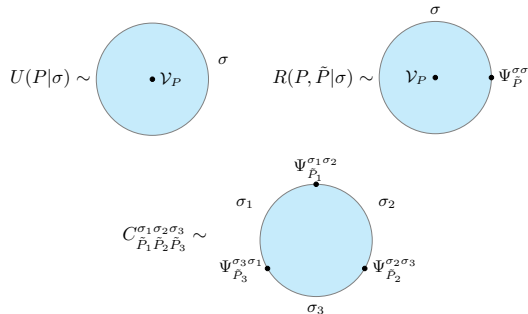
$$\mathcal{S}_B = \mu_b \int_{\partial M} e^{b\phi} d\zeta, \quad \mu_b^2 = \frac{\mu \cosh^2(\pi\sigma/b)}{\sin(\pi b^2)}.$$

Boundary insertions introduce new CFT data.

KNOWN IN THE SPACELIKE CASE

FZZT used bootstrap equations and Coulomb-gas/free-field techniques.

The analytic continuation to timelike Liouville is not straightforward.



STRATEGY

- > Merely a one-to-one correspondence between kernels and boundary CFT data
⇒ Crossing equations = consistency conditions of the Moore-Seiberg groupoid.
- > Recent progress concerning the derivation of the kernels of timelike Liouville for $\hat{b}^2 \in \mathbb{Q}$ via Virasoro Wick rotation.

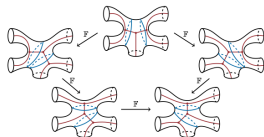
TIMELIKE CASE

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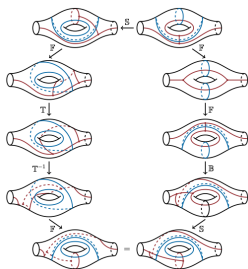
$$\begin{aligned} & \text{Diagram 1} = \int_0^\infty dP' \mathbb{F}_{P,P'} \begin{bmatrix} P_3 & P_2 \\ P_4 & P_1 \end{bmatrix} \text{Diagram 2} \\ & \text{Diagram 3} = \int_0^\infty dP_2 \mathbb{S}_{P_1,P_2}[P_0] \text{Diagram 4} \end{aligned}$$

RELATION BETWEEN CFT DATA AND KERNELS



Pentagon equation $\leftrightarrow$ Crossing 4 pts bdy

$$\int \mathbb{F}\mathbb{F}\mathbb{F} = \mathbb{F}\mathbb{F} \leftrightarrow \int \mathbb{C}\mathbb{C}\mathbb{F} = \mathbb{C}\mathbb{C}$$



Torus 2 pt $\leftrightarrow$ 2 bulk and 1 bdy crossing

$$\mathbb{S}\mathbb{F} = \int \int e^{i\theta} \mathbb{S}\mathbb{F}\mathbb{F}\mathbb{F} \leftrightarrow \mathbb{R}\mathbb{C} = \int \int e^{i\theta} \mathbb{R}\mathbb{C}\mathbb{F}\mathbb{F}$$

- > In the spacelike case, this gives $C \propto \mathbb{F}$, $R \propto \mathbb{S}$ with $\mathbb{F}$ and $\mathbb{S}$ known.
- > In the timelike case, it remains to find kernels solving the consistency relations.

WORKING AT RATIONAL $\hat{b}^2$

- The shift equations admit a two-dimensional vector space of kernel solutions.
- The physical kernel solving four-point crossing is non-meromorphic.
- The candidate CFT data are associated with the meromorphic Virasoro-Wick-rotated kernels.

CONJECTURAL IDENTIFICATION

$$C \propto \mathbb{R}\mathbb{F}, \quad R \propto \mathbb{R}\mathbb{S}.$$

Numerical evidence and analytic arguments support this construction, with evidence for uniqueness for $c \leq 1$.

CONCLUSIONS AND NEXT STEPS

WINEGLASS COSMOLOGY

Phase diagram completed; next step: finish the power spectrum analysis and sharpen the WKB matching argument.

BOUNDARY TIMELIKE LIOUVILLE

Candidate boundary CFT data from Virasoro-Wick-rotated kernels. Next step: write down the analytic derivation and produce precise numerical plots showing evidence for the conjecture.

OTHER DIRECTIONS

Entanglement entropy in $c = 1$ MQM; axion wormholes and dynamical-systems methods.

Near-term objective: complete the ongoing projects/prepare the corresponding papers/present the works next year.